

1. To run the program download both files into a python 3 IDE. Then run the chatServer.py file first and then run the chatClient.py file. You can run them by typing python 'filename.py' in the IDE terminal

The program works as a turn based chat system that also allows the users to play a rock paper scissors game. The conversation first starts with the client. Whoever starts the game makes the first selection of rock paper or scissors. Whichever the client or server that initiates the game also starts the next chat when the game ends. The program does not accept blank string ("") as input for the chat and tells the user to re-input a message. The program takes strings up to 4096 characters. The program prints responses except for '/q' and the '/rockpaperscissors' where it instead prints statements indicating the client/server shutting down or playing a game

2.

Client side

```
C:\Users\Vern\AppData\Local\Programs\Python\Python310\python.exe C:/Users/Vern/PycharmProjects/CS372/Project4/chatClient.py
connected to: 127.0.0.1 on port: 8001
type /q to quit
Enter message to send. Please wait for input prompt before entering message...
Enter /rockpaperscissors to play a game of rock, paper, scissors
Enter input >Hello
How are you?
Enter input >Good. You?
Good
Enter input >Bye
Bye
Enter input >/q
Shutting down
Process finished with exit code 0
```

Server side

```
C:\Users\Vern\AppData\Local\Programs\Python\Python310\python.exe C:/Users/Vern/PycharmProjects/CS372/Project4/chatServer.py
Server listening on: 127.0.0.1 port: 8001
Connected by: ('127.0.0.1', 61796)
Waiting for message...
Hello
type /q to quit
Enter /rockpaperscissors to play a game of rock, paper, scissors
Enter message to send. Please wait for input prompt before entering message...
Enter input >How are you?
Good. You?
Enter input >Good
Bye
Enter input >Bye
Client requested shutdown. Shutting down
Process finished with exit code 0
```

3. Extra Credit: Program has additional functionality to not accept an empty message:

```
while (message == ''):  
    print('An empty message is not valid. Please enter a valid message')  
    message = input(PROMPT)
```